HW 8:

Prob 1:

$V_{dc} = 10.74V$

$\Delta t = 23.5mS$

$\underline{I_d^{\max} = 4.915A}$

Prob 2:

$V_{z0} = 11.4V$

$V_z = 12.15V$

$R = 421.04\Omega$

$\underline{Line\_Regulation = 0.064V / V}$

Prob 3:

a) $V_{out} = \dfrac{V_{IN} - 0.7}{2}$

Prob 4:

$$V_{out} = \frac{V_{IN} - 0.7}{3}$$

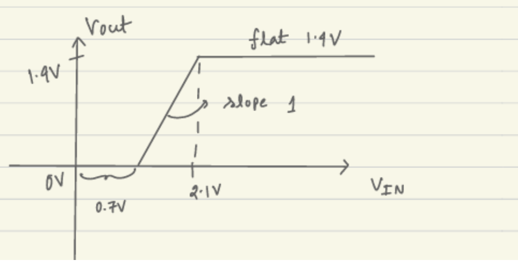
Prob 5:

on, Vout = Vin − 0.7V

and if D2 turns on then Vout ⩾ 1.4

∴ Vout = Vin − 0.7V

If, V_{IN} ⩾ 2.1V then, Vout = 1.4V



Prob 6:

a) $I_E = I_B + I_C = 1.01\,mA$

$\alpha = I_C / I_E = 0.99$

$\beta = I_C / I_B = 100$

$I_S = I_C / e^{V_{BE}/V_T} = 6.91 \times 10^{-16}\,A$

b) $I_E = I_B + I_C = 20\,mA$

$\alpha = I_C / I_E = 0.98$

$\beta = I_C / I_B = 50$

$I_S = I_C / e^{V_{BE}/V_T} = 1.031 \times 10^{-15}\,A$

c) $I_E = I_B + I_C = 0.23\,mA$

$\alpha = I_C / I_E = 0.978$

$\beta = I_C / I_B = 46$

$I_S = I_C / e^{V_{BE}/V_T} = 19.32 \times 10^{-15}\,A$

d) $I_E = I_B + I_C = 10.220\,mA$

$\alpha = I_C / I_E = 0.988$

$\beta = I_C / I_B = 84.167$

$I_S = I_C / e^{V_{BE}/V_T} = 2.847 \times 10^{-16}\,A$

e) $I_C = 73.95\,mA$

$\alpha = 0.986$

$\beta = 70.429$

$I_S = 4.208 \times 10^{-16}\,A$

Prob 7:

a) I1=2mA
   V2=0.7V
b) V3=-2V
c) I5=0.5mA
d) I6=1.59mA